

Economic Development Data Engineering Portfolio — Design Proposal

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Overview

This portfolio is designed to demonstrate production-grade data engineering skills through real, publicly available economic data rather than a generic tutorial dataset. Each project starts from a genuine economic development question, then builds the pipeline needed to answer it — ingestion, transformation (dbt), data quality testing, and an analytical output.

The portfolio is built to support interviews for Data Engineering and Analytics Engineering roles, including public-sector / government DE roles, where publicly sourced data, transparent methodology, and data quality rigor tend to matter more than flashy consumer-app projects.

Analysis and dashboarding will be done in **Jupyter Notebook** rather than a BI tool like Looker Studio, keeping the full pipeline — from raw ingestion through final analysis — in a reproducible, code-first format that's easy to walk through live in an interview.

Project 1: Growth vs. Development (Flagship)

Business question: Where is economic growth actually translating into broad-based development, versus concentrated gains? This compares GDP growth against development indicators (poverty rate, mobile penetration, employment) across a set of African economies, rather than relying on a single headline metric.

Countries: Nigeria, Kenya, Ghana, Rwanda, South Africa (small, deliberately scoped set)

Indicators: GDP growth, poverty headcount ratio, mobile subscriptions per 100 people, unemployment rate

Data source: World Bank Indicators API (`api.worldbank.org`) — free, no authentication required, returns JSON

Architecture

1. **Ingestion** — Python script pulling from the World Bank API on a schedule

2. **Raw storage** — land raw JSON responses into BigQuery as-is (bronze/raw layer pattern)
3. **Transformation (dbt)**
 - Staging models: flatten raw JSON into tidy rows (country-indicator-year)
 - Intermediate models: pivot indicators into wide format
 - Mart model: final analysis-ready table joining growth and development indicators by country-year
 - dbt tests: not-null checks, accepted country codes, reasonable value ranges
4. **Orchestration** — Airflow DAG wrapping ingestion + dbt run, scheduled to refresh periodically (phase two, once the core pipeline works)
5. **Analysis / dashboard** — Jupyter Notebook reading from BigQuery, with charts and a written walkthrough answering the business question directly; kept in the same repo as the pipeline so the whole thing — ingestion through analysis — is reproducible and easy to demo live in an interview

Cost

Essentially **\$0/month**. BigQuery's free tier includes the first 1 TiB of query data processed per month and the first 10 GB of storage free. A dataset this size (a few thousand country-year rows) is a few MB, and even frequent dbt runs stay nowhere near the free query threshold.

Timeline

2-3 weeks at a part-time pace — ingestion script, raw + staging + mart dbt models, tests, and the Jupyter analysis notebook. Orchestration (Airflow) can follow as a phase two once the core pipeline is working and demoable.

Project 2: FDI vs. Growth

Business question: Which countries or sectors show the strongest link between foreign direct investment (FDI) and long-term GDP growth? FDI doesn't always translate into growth, so this has real analytical teeth rather than being a single chart.

Data sources: World Bank FDI indicators + UN Comtrade (trade/investment flow data)

Why it's next: the ingestion → dbt → notebook pattern from Project 1 is already built, so this is mainly about adding a new source and a proper dimensional model — multiple fact tables (investment flows, GDP) around shared dimensions (country, year, sector).

Timeline: roughly 1.5–2 weeks — faster than Project 1 since the pipeline pattern is already established.

Project 3: Mobile Money / Fintech Adoption

Business question: How has mobile money / fintech adoption correlated with financial inclusion and small-business growth across African markets? A timely, specific angle that demonstrates awareness of current economic development trends, not just generic macro indicators.

Data sources: Mobile money / fintech adoption data (e.g. GSMA, World Bank Findex), cross-referenced with small-business growth indicators

Why it's last: this data is messier and less standardized than GDP or FDI data, so it's the best justification in the whole portfolio for real data-quality and validation work — dbt tests earning their keep, not just boilerplate.

Timeline: roughly 2–2.5 weeks — data cleaning is the bottleneck here, not pipeline architecture.

Overall Timeline

Phase	Scope	Estimate
Project 1 — Growth vs. Development	Ingestion → dbt staging/marts → tests → Jupyter analysis	2–3 weeks
Project 2 — FDI vs. Growth	New source, new fact table, extended analysis	1.5–2 weeks
Project 3 — Mobile Money / Fintech	Messier source, heavier data-quality work	2–2.5 weeks
Polish pass	README, architecture diagram, question-to-answer writeup per project, portfolio presentation	3–5 days

Total: roughly 7–9 weeks at a part-time pace, alongside Le Wagon prep and job searching — not full-time bootcamp hours.

Project 1 alone, finished well with a working notebook and a clear question-to-answer writeup, is enough to start showing in interviews. Projects 2 and 3 can be built *while* job searching — “currently expanding this to cover FDI” is a legitimate, honest thing to say live in an interview.

Cost Summary

All three projects can run on BigQuery's free tier at essentially **\$0/month**:

- First 1 TiB of query data processed per month: free
- First 10 GB of storage: free
- Dataset size across all three projects (World Bank, UN Comtrade, mobile money indicators for a handful of countries) is a few MB — well under either threshold

No cloud spend is expected to build or maintain this portfolio.

Positioning for Government DE Interviews

Government and public-sector data engineering roles tend to value this kind of project more than a flashy consumer-app clone, because it demonstrates:

- **Public data sources** — World Bank, UN Comtrade, GSMA/Findex — the same category of source many public-sector roles work with directly
- **Transparent methodology** — every transformation step is visible and reproducible in dbt + notebook form
- **Data quality rigor** — dbt tests on messy, real-world data (especially in Project 3) mirror the data-quality expectations common in public-sector pipelines
- **A policy-relevant question** — each project starts from a genuine economic development question rather than an arbitrary technical exercise

If targeting Canadian government or Quebec-specific roles specifically, a smaller fourth mini-project using Statistics Canada or Quebec public data could reinforce this positioning further, once the three flagship projects are done.